Radio Practical Skills Trainer Guide

1-2 trainers, 90-120 Minutes

Print the Following Documents

Three (3) full, double-sided pages per person, and one (1) double-sided cheat sheet.

- *Double sided bingo sheets for each participant*
- *Half-page cheat sheet for each participant*
- *Two sets of double-sided handouts for each <u>pair of</u> participants*
    - *This is actually two separate, double-sided handouts*

Introduction

- Welcome!
- Participants give name, pronouns, one reason they're here.
- Trainer intros.
- No experience necessary or expected.
- We will give out handouts – you don't have to take notes!
- By the end of this training, you will have the skills you need to practice using FRS with your friends / neighbors / affinity group to prepare for ICE raids or a major disaster.
- Who here brought an FRS radio? Can you hold it up?

What is FRS

- What is FRS – and how is it practically different to other common radio systems?
- CB: "trucker radio". Somewhat a victim of its own popularity and lack of regulation/coordination/enforcement.
- FRS: Most are cheap and not recommended: Low power, fixed antenna, cheaply made.
- But for our purposes they're great, bc we're going to be close to each other.
- GMRS: Growing in popularity, a great gateway to approachable practical radio usage. Licensed required but not enforced for most uses
- Ham (amateur): Surging in popularity, both practical uses and as a hobby - higher learning curve/knowledge testing, but potentially to talk around the world

Radio & FRS Basics

- Radio waves are divided up into sections – called "bands" – for different purposes.
    - For example, the 5 gigahertz band is allocated to 5G WiFi.
- There is a medium-sized section allocated to amateur (or Ham) radio.
    - That slice allows for hundreds of frequencies that a Ham radio operator can use.
- FRS (and GMRS) gets a pretty narrow section of the spectrum: just 22 channels.
    - What's a channel? We'll come back to that.
- FRS operates on what is called "Line of Sight". Physical obstructions, especially metal (buildings) and dirt (hills, etc.) drastically limit its range.
- Practical range? A half mile or less in an urban setting.

Frequencies and Channels

- Frequencies are what a radio transmits/receives on.
- FRS'es are channelized – the frequencies for all 22 channels are fixed by the FCC.
- Changing transmission frequencies is illegal, and basically impossible to do.
- All FRS radios share the same 22 channels: Channel 17 is the same on every modern FRS radio, and every GMRS, too.
- What does this mean? Everyone shares these limited channels, and you'll often find other people using FRS channels. As it gains popularity, it gets crowded!
- To hear someone, you must be receiving on the same channel they are transmitting on.
- Be sure to set up a backup channel if yours is "taken" or ends up being used.

Radio Etiquette

- Every FRS channel is shared by every FRS radio user. (within range)
    - Every minute you speak is a minute that one or more other people – maybe many more people! – can't use the channel.
- To make up for these limitations, radio operators have developed "radio etiquette".
    - These are guidelines to make radio communications more efficient and effective.
- Here are the very basics of radio etiquette:
    - *Hand out the radio cheat sheet.*
    - Now take a moment to read through the radio etiquette. Any questions?
- *Demonstrate using the NATO alphabet by spelling your name with it, etc.*

Radio Bingo

- You're going to have a chance to practice radio etiquette and some common radio lingo.
    - *Pass out Radio Bingo handout.*
    - *Give participants a minute to read the terms and how they're used.*
    - *Trainers do a short demonstration of using radio etiquette to play radio bingo.*
- Take a moment to read the instructions and the glossary on the back. Any questions?
    - *Participants pair up for Radio Bingo.*
    - *They go for ~10 minutes or until people have gotten comfortable using etiquette.*
- How did that go? Did it become a bit more natural to use radio etiquette and lingo?

Trainer's Note

- During this activity, use your FRS to monitor the channels participants are using.
- Get a sense of how well they are doing in real time.
    - If anyone is really struggling (not holding the PTT down the whole time they're talking, etc.) intervene right away so they get the most of the practice time.
    - Also feel free to walk up to participants to help them fix any HT problems, etc.
- At the end, say "(Your name), break, break, break, priority. Bingo is finished. Come back and tell anyone you see on the way."
    - This is a fun way to use radio terminology in an organic way.

Radio Communication Practice

- The best way to learn to use the radio is getting on the air.
- This next exercise will get you practice passing information to each other.
- These messages are intentionally goofy instead of serious so that, if they are overheard, people won't think that there are, say, emergencies actually happening.
    - But you will be able to use the exact same skills you will learn today to share information about actually important things.
- Now break into partners!
    - *Pass out the first worksheets. Each partner gets a different one, either A or B.*
    - Take a moment to read this *to yourself*, while keeping it secret from your partner.
    - The messages are numbered one to five, and increase in difficulty as you go.
        - Do you have any questions?
    - First person A reads their number 1, and then person B reads their number 1.
    - Write down what you think is the important information in each message.
    - When you're both done, get together and evaluate your partner's performance.
- *Have partners count off from one. Tell them that's the channel number they will use.*
- Now split up and walk far enough from your partner that you can't hear each other except through the radio.
    - But not so far it'll take too long to walk back when you're done.

Troubleshooting

- In some enviroments, different groups will hear transmissions from adjacent channels.
- If that happens, have participants use channels that are at least two apart.
- In extreme cases you can have participants use channels that are three apart.
- In the debrief, explain that cheaper radios can transmit and receive outside the channels they're programmed for, and this is something you can learn and work around when practicing with your own equipment.

Radio Communication 1 Evaluation

- How did things go?
- What was hard? What kind of errors did you make or see?
    - Homophones like *to*, *too*, and *two*.
    - Proper names and street addresses.
    - Technical terms like "Azithromycin" and "dihydrogen monoxide". (water)
    - Email addresses with a mix of letters and symbols.
    - When it's unclear if a sentence is a question or not.
    - Writing with good handwriting, quickly!
- What are some ways you can make it easier to communicate over the radio?
    - Repeat back what your partner says to you
    - Use radio etiquette, speak slowly and clearly, use the NATO alphabet…
    - You can craft your message to be as clear and unambiguous as possible.
        - Use synonyms that are easier to understand on the radio.
    - Say "email address" before reading one off, to prepare the listener.
        - You can say "building name" or "medicine name" the same way.

    - When needed, say punctuation out loud, like “question mark”.
- The listener is an equal participant in the communication!
    - When needed, ask the speaker to go more slowly, etc.
    - If need be, specify what part of the message needs to be repeated, so they don’t repeat the whole thing.
- Note: There are trainings specifically on message passing, which is copying down long messages, letter-for-letter perfect; no paraphrasing!
    - It’s hard but also very rewarding.

Evaluation

- What did folks like about this workshop?
- What can we improve for next time?
- *Reminder to trainers: Never disagree or argue with participant feedback!*